NAME OF THE FACULTY: S. ALIVELU

Lesson Plan: Programming in C (CM-106) Academic Year: 2025-26

UNIT 1: Introduction to C Language (20 Periods)				
S.No	**Date**	**Name of the topic to be covered**	**No. Of Periods**	**Completed Date**
1.	14/07/2025	History of C-language,importance of C language.	1	
2.	15/07/2025	Structure of a C-language program, programming style.	1	
3.	16/07/2025	Steps in executing a C program (Editing,Compiling)	1	
	17/07/2025	**Muharram – HOLIDAY**		
4.	18/07/2025	Steps in executing a C program (Executing and Debugging).	1	
5.	19/07/2025	C Character set, C-Tokens (keywords, identifiers).	1	
6.	21/07/2025	C-Tokens (Constants,Variables).	1	
7.	22/07/2025	Data Types and classification - Primitive data types.	1	
8.	23/07/2025	Declaration of constants and variables.	1	
9.	24/07/2025	Initializing values to variables.	1	
10.	25/07/2025	User defined data types	1	
11.	26/07/2025	Usage of type qualifiers (const, volatile).	1	
12.	28/07/2025	Introduction to Pre-processor Directives, Usage of #include.	1	
13.	29/07/2025	Reading a character using getch() and getche ().	1	
14.	30/07/2025	Reading a character using getchar().	1	
15.	31/07/2025	Writing a character using putch() and putchar ().	1	
16.	01/08/2025	Formatted input using scanf() with sample programs.	1	
17.	02/08/2025	Formatted output using printf() with sample programs	1	
18.	04/08/2025	Introduction to character functions.	1	
19.	05/08/2025	Revision of Unit 1 Topics.	1	
20.	06/08/2025	Problem solving session for Unit 1.	1	
(UNIT 1 END)			**Total 20 Periods**	

UNIT 2: Input and output statements, Operators and Expressions in C (25 Periods)				
S.No	**Date**	**Name of the topic to be covered**	**No. Of Periods**	**Completed Date**
1	07/08/2025	Introduction to Unit 2: Definition of Operators and Expressions	1	
2	08/08/2025	Various Arithmetic Operators (+, -, *, /, %)	1	
3	09/08/2025	Evaluation of Arithmetic Expressions with examples	1	
4	11/08/2025	Relational Operators (==, !=, >, <, >=, <=)	1	
5	12/08/2025	Evaluation of Relational Expressions	1	
6	13/08/2025	Logical Operators (&&, ||, !) and Truth Tables	1	
7	14/08/2025	Evaluation of Logical Expressions & Short-circuiting	1	
--	15/08/2025	Independence Day – HOLIDAY	--	
8	16/08/2025	Difference between Unary and Binary operators	1	
9	18/08/2025	Increment (++) and Decrement (--) Operators (Prefix vs Postfix)	1	
10	19/08/2025	Simple and Compound Assignment Operators (+=, -=, *=, /=)	1	
11	20/08/2025	Illustrating Nested Assignment (a = b = c = 10)	1	
12	21/08/2025	Conditional (Ternary) Operator (? :) with examples	1	
13	22/08/2025	Bit-wise AND (&), OR (|), and XOR (^)	1	
14	23/08/2025	Bit-wise NOT (~) and Shift Operators (<<, >>)	1	
15	25/08/2025	Special Operators: sizeof(), comma (,), and address (&)	1	
16	26/08/2025	Operator Precedence and Associativity Rules	1	
17	27/08/2025	Solving the Operator Precedence Table (The "BODMAS" of C)	1	
18	28/08/2025	Evaluation of Compound Expressions	1	
19	29/08/2025	Implicit Type Conversion (Coercion)	1	
20	30/08/2025	Explicit Type Conversion (Type Casting)	1	
21	01/09/2025	Sample Program 1: Arithmetic and Logical Calculator	1	
22	02/09/2025	Sample Program 2: Bitwise manipulation and Ternary Logic	1	
23	03/09/2025	Lab Practice: Debugging Operator-based errors	1	
24	04/09/2025	Revision of Unit 2: Key Concepts & Shortcuts	1	
25	05/09/2025	Unit 2 Assessment / Quiz	1	
		(UNIT 2 END)		

UNIT 3: Decision making, iterative and other control statements (40 Periods)				
S.No	**Date**	**Name of the topic to be covered**	**No. Of Periods**	**Completed Date**
1	08/09/2025	Introduction to Decision Making & its Need in C	1	
2	09/09/2025	Simple if Statement: Syntax and Logic	1	
3	10/09/2025	Simple if Statement: Sample Programs	1	
4	11/09/2025	if-else Statement: Syntax and Logic	1	
5	12/09/2025	if-else Statement: Sample Programs	1	
6	13/09/2025	Lab Practice: Programs on if and if-else	1	
7	15/09/2025	Nested if...else Statements: Syntax and Logic	1	
8	16/09/2025	Nested if...else: Sample Programs	1	
9	17/09/2025	if-else-if Ladder: Syntax and Logic	1	
10	18/09/2025	if-else-if Ladder: Sample Programs (Grades/Marks)	1	
11	19/09/2025	switch Statement: Syntax and Rules	1	
12	20/09/2025	Importance of break in switch (Fall-through)	1	
13	22/09/2025	Programming Practice: Menu-driven Calculator	1	
14	23/09/2025	Comparison: Conditional Operator vs if-else	1	
15	24/09/2025	Comparison: if-else vs switch Statement	1	
16	25/09/2025	Review & Problem Solving: Decision Making	1	
17	26/09/2025	Introduction to Looping (Iteration)	1	
18	27/09/2025	The while Loop: Syntax and Execution Flow	1	
19	29/09/2025	The while Loop: Sample Programs (Counting/Summing)	1	
20	30/09/2025	The do-while Loop: Syntax and Execution Flow	1	
21	01/10/2025	The do-while Loop: Sample Programs	1	
--	02/10/2025	Gandhi Jayanthi – HOLIDAY	--	
22	03/10/2025	Comparison: while vs do-while Loops	1	
23	04/10/2025	The for Loop: Syntax and 3-expression structure	1	
24	06/10/2025	The for Loop: Sample Programs (Factorial/Tables)	1	
25	07/10/2025	Advanced for Loop: Multiple Init/Increment	1	
26	08/10/2025	Comparison: while, do-while, and for loops	1	
27	09/10/2025	Nested Loops: Definition and Logic	1	
28	10/10/2025	Illustrating Nested Loops with Table Printing	1	
29	11/10/2025	Pattern Printing using Nested Loops (Part 1: Basic)	1	
30	13/10/2025	Pattern Printing using Nested Loops (Part 2: Advanced)	1	
31	14/10/2025	Usage of break statement with loops	1	
32	15/10/2025	Usage of continue statement with loops	1	
33	16/10/2025	Usage of goto statement and Labels	1	
34	17/10/2025	Differentiate break and continue with examples	1	
35	18/10/2025	Problem Solving: Loop Jump Scenarios	1	
36	20/10/2025	Infinite Loops: Causes and How to Avoid Them	1	
37	21/10/2025	Definition and Importance of Structured Programming	1	
38	22/10/2025	Comprehensive Program: Combining Decison & Loops	1	
39	23/10/2025	Unit 3 Revision: Core Syntax & Logic	1	
40	24/10/2025	Unit 3 Assessment / Test	1	
		(UNIT 3 END)	**Total 40 Periods**	

UNIT 4: Arrays and strings, Structures and Unions (30 Periods)				
S.No	**Date**	**Name of the topic to be covered**	**No. Of Periods**	**Completed Date**
1	27/10/2025	Introduction to Arrays: Definition and Concept	1	
2	28/10/2025	Declaration and Initialization of 1D-Arrays	1	
3	29/10/2025	Accessing 1D-Array Elements (Input/Output)	1	
4	30/10/2025	Programming: Finding Sum and Average of Array Elements	1	
5	31/10/2025	Reordering an Array: Sorting in Ascending Order	1	
6	01/11/2025	Lab Practice: 1D Array manipulation and Sorting	1	
7	03/11/2025	Introduction to 2D Arrays: Declaration and Initialization	1	
8	04/11/2025	Accessing Elements in 2D Arrays (Row & Column logic)	1	
9	05/11/2025	Matrix Operations: Addition and Subtraction	1	
10	06/11/2025	Matrix Operations: Multiplication Logic	1	
11	07/11/2025	Programming: Implementing Matrix Multiplication	1	
12	08/11/2025	Introduction to Strings: Definition and Memory Layout	1	
13	10/11/2025	String Declaration and Initialization (Null Character)	1	
14	11/11/2025	Reading/Writing Strings: gets() and puts()	1	
15	12/11/2025	String Handling Functions: strlen(), strcpy()	1	
16	13/11/2025	String Handling Functions: strcat(), strcmp()	1	
17	14/11/2025	Character Arithmetic (ASCII logic with strings)	1	
18	15/11/2025	Lab Practice: String manipulation without library functions	1	
19	17/11/2025	Introduction to Structures: Definition and Need	1	
20	18/11/2025	Declaring and Initializing Structure Variables	1	
21	19/11/2025	Accessing Structure Members (Dot Operator)	1	
22	20/11/2025	Structure Assignment and Finding Size (sizeof)	1	
23	21/11/2025	Nested Structures: Concept and Syntax	1	
24	22/11/2025	Structures containing Arrays	1	
25	24/11/2025	Array of Structures (Storing multiple records)	1	
26	25/11/2025	Introduction to Unions: Definition and Declaration	1	
27	26/11/2025	Accessing Union members and Memory Allocation	1	
28	27/11/2025	Distinguish between Structures and Unions	1	
29	28/11/2025	Programming Practice: Practical use cases for Unions	1	
30	29/11/2025	Unit 4 Review and Assessment	1	
		(UNIT 4 END)	**Total**	**30 Periods**

UNIT 5: User defined functions, pointers, file management and pre-processor directives (35 Periods)				
S.No	**Date**	**Name of the topic to be covered**	**No. Of Periods**	**Completed Date**
1	02/12/2025	Need and Advantages of User-Defined Functions	1	
2	03/12/2025	Elements of Functions: Declaration, Definition, & Call	1	
3	04/12/2025	Return Values, Types, and Function Prototypes	1	
4	05/12/2025	Functions: No arguments & No return values	1	
5	06/12/2025	Functions: With arguments & No return values	1	
6	08/12/2025	Functions: With arguments & With return values	1	
7	09/12/2025	Functions returning multiple values (Logic)	1	
8	10/12/2025	Introduction to Recursion: Concept and Memory Stack	1	
9	11/12/2025	Programming Practice: Factorial and Fibonacci via Recursion	1	
10	12/12/2025	Passing Arrays to Functions with sample programs	1	
11	13/12/2025	Structures as Function Arguments and Return Values	1	
12	15/12/2025	Storage Classes: auto, register, static, and extern	1	
13	16/12/2025	Scope, Visibility, and Lifetime of variables	1	
14	17/12/2025	Local vs External variables & Global variables	1	
15	18/12/2025	Introduction to Pointers: Declaration & Initialization	1	
16	19/12/2025	Accessing Address (&) and Value through Pointer (*)	1	
17	20/12/2025	Pointer Arithmetic and Operator Precedence	1	
18	22/12/2025	Relationship between Arrays and Pointers	1	
19	23/12/2025	Accessing Array elements using Pointers	1	
20	24/12/2025	Pointers as Function Arguments (Call by Reference)	1	
--	25/12/2025	Christmas – HOLIDAY	--	
21	26/12/2025	Pointer Arrays with practical examples	1	
22	27/12/2025	Dynamic Memory Allocation: malloc(), calloc()	1	
23	29/12/2025	Dynamic Memory Allocation: realloc(), free()	1	
24	30/12/2025	Pointers to Structures & Self-referential structures	1	
25	31/12/2025	Introduction to Files: File Pointers and fopen()	1	
26	01/01/2026	File Opening Modes (r, w, a, r+, w+, a+)	1	
27	02/01/2026	Closing Files and Basic I/O: fgetc(), fputc()	1	
28	03/01/2026	Formatted File I/O: fprintf() and fscanf()	1	
29	05/01/2026	Random Access Files: fseek(), ftell(), rewind()	1	
30	06/01/2026	Different File Handling functions (Review)	1	
31	07/01/2026	Preprocessor Directives: Need and Types (#define, #if)	1	
32	08/01/2026	Simple programs using Macros and Directives	1	
33	09/01/2026	Command Line Arguments: argc and argv logic	1	
34	10/01/2026	Comprehensive Lab: Student File Management System	1	
35	12/01/2026	Final Revision and Unit 5 Assessment	1	
	(UNIT 5 END)		**Total : 35 Periods**	